

Rule-Guided Weak Supervision for Entity-Aware Aviation Speech Recognition and Unified Text Restoration

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Abstract

Operational entities in aviation speech often have markedly different spoken and written realizations, making entity preservation during automatic speech recognition (ASR) critical to downstream canonicalization. We present a schema-guided framework that reuses aviation structure at training and inference. During training, deterministic matches on spoken-form references provide token-level weak entity labels for an auxiliary objective during LoRA adaptation. During inference, a schema-constrained restoration module converts only validated entity patterns and copies unsupported spans unchanged. We evaluate the framework on a leakage-audited dataset of 58,507 controlled aviation-English course recordings with paired prompt-derived spoken-form and canonical references. On 4,487 test recordings, auxiliary entity supervision reduces WER against the spoken-form reference from 0.5593 to 0.5255 relative to matched LoRA, a 6.0% relative reduction (Holm-adjusted $p = 0.029$). For the same recognizer outputs, the restoration module reduces canonical WER from 0.6618 with fixed aviation ITN to 0.5857, an 11.5% relative reduction (Holm-adjusted $p < 0.001$), and increases normalized sentence exact match from 28.91% to 32.85%. These results support schema sharing across recognition and canonicalization, while entity-level analysis shows that aggregate WER gains do not imply uniform improvements across entity types.

CCS Concepts

• **Computing methodologies** → **Speech recognition**; *Information extraction*; *Multi-task learning*.

Keywords

aviation speech recognition, weak supervision, entity-aware ASR, text restoration, inverse text normalization

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1 Introduction

Air-ground communication relies on concise and highly structured language. A single utterance may combine ordinary English with call signs, runway designators, headings, flight levels, frequencies, and other operational values. ASR systems in this domain must transcribe ordinary lexical content while preserving these entities, for which even a single-token error can alter an instruction. Existing benchmarks document difficulties arising from channel conditions, accents, specialized phraseology, and limited in-domain supervision [34, 35].

The text required by downstream applications also differs from a literal spoken-form transcript. For example, *runway zero two right* may be represented as *RWY 02R*, and *heading zero tree zero*

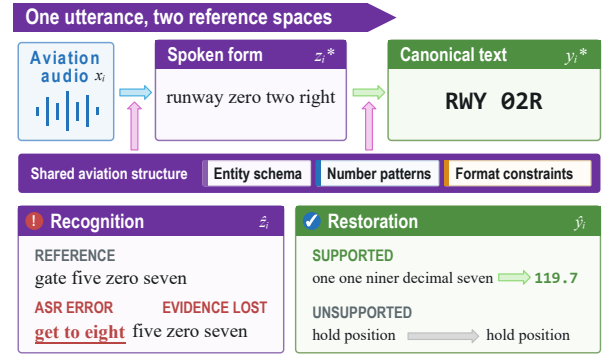


Figure 1: Motivation for entity-preserving recognition and constrained canonicalization. An ASR error may remove the evidence required to recover an aviation entity; restoration therefore converts only schema-validated spans and copies unsupported spans unchanged.

as heading 030. Figure 1 illustrates two coupled challenges. An ASR substitution can remove evidence needed downstream, while a canonicalization module should transform a span only when the hypothesis provides sufficient lexical and structural evidence. General ITN addresses spoken-to-written conversion [28, 33], but aviation communication also requires entity-specific formats and conservative copy-through behavior.

Domain adaptation and contextual biasing offer complementary approaches to specialized ASR. LoRA can specialize a pretrained model with few trainable parameters [10, 17]. Contextual methods favor domain terms through an external vocabulary or biasing signal [25, 26], but depend on candidate coverage and relevance at inference. Rule-based post-processing avoids this dependency during decoding, yet cannot recover entity evidence already deleted or substituted by ASR. Spoken entity recognition connects transcription and entity extraction [1], but usually assumes entity supervision or a task-specific output format.

We propose a rule-guided weak-supervision framework that connects recognition and restoration through a shared aviation schema. Deterministic matches on training references produce token-aligned entity labels, and an auxiliary objective encourages the adapted decoder to retain entity-bearing evidence. The classifier is removed after training and does not change the transcript space. At inference, a schema-constrained restoration module detects supported patterns, composes canonical forms, validates local constraints, and otherwise copies the input unchanged. The primary task maps aviation audio to canonical text; comparison with the expected spoken form diagnoses recognition.

We construct paired spoken-form and canonical references from structured course recordings. Audio deduplication and grouped splitting keep audio hashes, normalized text, and dialogue identities

disjoint across partitions. Under the controlled evaluation, auxiliary entity supervision reduces WER against the prompt-derived spoken-form reference, while constrained restoration improves canonical generation over fixed aviation ITN. Because the data are prompted course recordings, the conclusions do not establish performance on live radio traffic.

Our contributions are threefold:

- We formulate aviation audio-to-canonical-text generation with paired prompt-derived spoken-form and canonical references, allowing recognition and canonicalization errors to be evaluated separately.
- We introduce schema-guided weak supervision for entity-preserving ASR adaptation without changing the inference-time transcript space or using a closed inventory of entity values.
- We reuse the same schema in a constrained restoration module and provide utterance-, entity-, and canonical-text analyses that separate the contributions of recognition adaptation and restoration.

2 Related Work

2.1 Aviation Speech Corpora and Recognition

ATC corpora span clean non-prompted controller speech, multilingual operational speech, and larger-scale recognition and language-understanding material [9, 32, 35]. The ATC benchmark measures general-to-domain transfer [34], while recent ATCCaps work evaluates call-sign-aware recognition on real ATC speech using manual transcripts and seen/unseen call-sign analyses [13]. Our controlled course setting instead studies paired spoken and canonical references together with conservative restoration.

ATC systems have incorporated operational context, model combination, and domain-specific adaptation [5, 18, 30]. Recent work applies LoRA to aviation transcription and analyzes acoustic, lexical, and speaker-role effects [3, 4, 17]. These studies improve transcription or characterize its errors; our focus is the connection between structure-supervised decoder adaptation and canonical restoration.

2.2 Contextual and Entity-Aware Speech Recognition

Contextual ASR injects external phrase information into end-to-end recognition. Deep Context conditions decoding on candidate phrases, while pointer-based models copy from structured lists or trees [14, 21, 27]. Other approaches align acoustic states with context terms through spike-triggered or intermediate biasing losses, and dynamic vocabularies extend this strategy to open Whisper-style models [11, 25, 26]. These methods are effective when reliable candidates are available, but a fixed global list can omit open-valued aviation compositions or over-bias decoding.

Spoken NER jointly predicts transcripts and entities through end-to-end or attention-based models [7, 20, 31]. Unpaired training reduces aligned-data needs, while WhisperNER unifies open named-entity and speech recognition [1, 16]. Our method retains an ordinary spoken-form transcript, derives labels only from training-side structure, and removes the auxiliary entity head after adaptation.

Table 1: Recording counts in the leakage-audited splits. All controlled adaptations use the same 48,098-recording training split (Section 6.1).

Split	Recordings
Training	48,098
Development	5,922
Test	4,487
Total	58,507

2.3 Weak Supervision for Structured Labels

Programmatic weak supervision replaces exhaustive manual labels with labeling functions and combines their outputs into training targets [24]. For NER, it can construct labels without a fully annotated corpus and train representations from noisy entity signals [15, 19]. Our setting differs in granularity: deterministic aviation rules first locate compositional spans in spoken-form training references, then align those spans with decoder tokens to provide a training-only representation signal.

2.4 Inverse Text Normalization and Aviation Restoration

ITN converts spoken expressions to written forms using data-driven grammars, weighted finite-state systems, or neural paired-text models [22, 28, 33]. Related systems jointly model ITN, punctuation, capitalization, and disfluency, or optimize normalization for streaming and on-device inference [6, 29]. Air-traffic command-center transcription has also motivated a dedicated ITN pipeline for domain formats [8].

Our task is instead defined from audio to canonical aviation text. Recognition errors determine which structures remain recoverable, while unconstrained normalization can alter otherwise valid hypotheses. We therefore use ITN as one operation within a schema-constrained restoration module. Entity detection, canonical composition, format validation, and conservative fallback share the aviation schema used for training supervision.

3 Dataset Construction

We construct a paired aviation-speech dataset for evaluating spoken-form recognition and canonical text restoration. The recordings were produced as part of structured aviation-English course exercises and were made available to the authors for research analysis by an independent third-party provider. Our modeling pipeline uses course metadata, target texts, pseudonymous user and recording identifiers, and audio; the source audio is not redistributed with this submission. The recordings comprise reading and simulated communication exercises rather than live operational radio traffic, so our conclusions are limited to this controlled setting.

3.1 Paired Reference Construction

Each recording is associated with two text targets. The canonical reference y_i^* is derived from the written course prompt and retains aviation conventions such as call-sign codes, runway designators, and formatted numerical values. The spoken-form reference z_i^* is

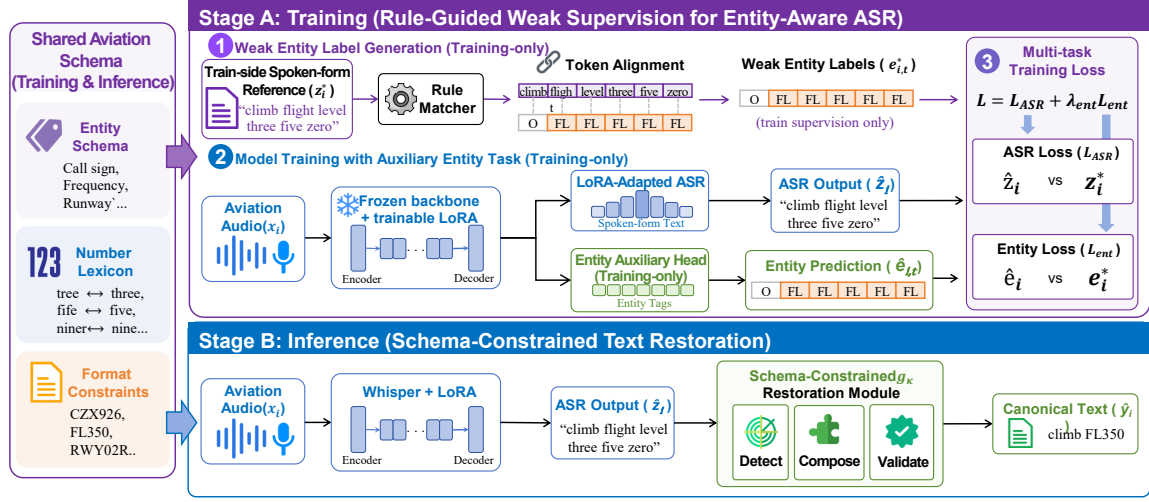


Figure 2: Overview of the schema-guided framework. During training, schema matches provide weak entity labels for a joint ASR and entity objective. During inference, the auxiliary classifier is removed and a schema-constrained restoration module maps the hypothesis to canonical text.

generated from the same prompt by the course phrase converter, which expands written structures into the word sequence that learners are instructed to pronounce. Thus, z_i^* is a prompt-derived spoken reference rather than a manually produced verbatim transcription of each recording. This paired construction supports separate evaluation of acoustic recognition against z_i^* and complete audio-to-canonical restoration against y_i^* .

The audio-bearing material covers four course-derived scenarios: word or phrase reading, example-sentence reading, single-turn scenario practice, and multi-turn communication training. These categories are inherited from the course organization and are not inferred from model performance.

3.2 Cleaning and Leakage-Audited Splits

We discard records with missing audio, missing target text, or unresolved audio–metadata correspondence. Duplicate audio is identified by content hash. Spoken and canonical references are normalized separately for matching and audit while preserving their distinct target conventions. Because multiple learners may record the same prompt, a random recording-level split would allow identical text to occur across partitions. We instead group examples by normalized text and dialogue identity before splitting, then verify that audio hashes and normalized text groups are disjoint across training, development, and test sets. Pseudonymous user identifiers are not used as split keys and are not validated as one-to-one speaker identities; the benchmark is therefore not intended to establish speaker-independent generalization.

Table 1 summarizes the split used throughout the paper. Development data are used for model and restoration choices. No development or test reference is used to create an entity pattern, phrase correction, or entity-value mapping; the exploratory adapter analysis is separately disclosed in Section 6.3.

4 Task Formulation

Given an aviation communication recording x_i , the system first predicts the prompt-derived spoken form expected by the course and then restores that form to canonical written text. Using the paired references introduced in Section 3, z_i^* is the spoken-form target and y_i^* is the canonical target. They describe related content but serve different purposes: z_i^* may contain *china star nine two six*, *runway zero two right*, whereas y_i^* contains CZX926, RWY 02R.

The ASR model f_θ predicts the spoken form

$$\hat{z}_i = f_\theta(x_i), \quad (1)$$

which is compared with z_i^* to measure spoken-form recognition. The complete system applies a restoration function g_K :

$$\hat{y}_i = g_K(\hat{z}_i) = g_K(f_\theta(x_i)). \quad (2)$$

The primary task compares $\hat{y}_i$ with y_i^* , while the spoken-form comparison diagnoses the upstream recognition capability.

We report corpus word error rate (WER) and normalized sentence exact match in both reference spaces. For the spoken-form output, a fixed aviation schema parser maps recognized entity spans to canonical (type, value) pairs; micro-averaged precision, recall, and F1 then measure whether call signs, runways, headings, altitudes, frequencies, and related structures are preserved. This entity-value score is computed on the reference-entity-positive subset and is a parser-based structural diagnostic, not a claim of manually annotated named-entity ground truth. We additionally report restoration regressions, defined as cases where an input already matching the canonical reference is changed into an incorrect output.

5 Method

5.1 Framework Overview

Figure 2 depicts the core framework. The central design is to reuse the same aviation structure across two modular stages: recognition learning and canonical restoration. We write this structure

as $\mathcal{K} = (C, \mathcal{L}, \mathcal{R})$, where C is the set of aviation entity types, $\mathcal{L}$ contains lexical variants for numbers and designators, and $\mathcal{R}$ contains compositional patterns and format constraints. The auxiliary label space is $C_0 = C \cup \{O\}$, where O denotes a non-entity token. As shown in Figure 2, the same interface generates training-only token labels and constrains canonical restoration. It represents open-valued structures rather than an inventory of every possible entity value, so decoding is not restricted to a fixed candidate list.

5.2 Training-Only Entity Weak Labels

The ordinary ASR objective treats an entity token like any other lexical token and does not explicitly identify spans whose internal digit sequence or designator structure must be preserved. We therefore use $\mathcal{K}$ to add weak supervision on the training side. Let $r_{\mathcal{K}}$ be a deterministic matcher over a spoken-form training reference z_i^* . It recognizes compositional patterns rather than memorizing complete entity values and returns

$$r_{\mathcal{K}}(z_i^*) = \mathcal{S}_i = \{(a_{i,k}, b_{i,k}, c_{i,k}, v_{i,k})\}_{k=1}^{K_i}, \quad (3)$$

where $K_i = |\mathcal{S}_i|$, $[a_{i,k}, b_{i,k})$ is a character span, $c_{i,k} \in C$ is its entity type, and $v_{i,k}$ is an optional canonical hint used to audit a matched span. The pattern set covers call signs, runways, headings, altitudes, flight levels, frequencies, QNH values, times, distances, passenger counts, squawk codes, and related numerical structures. Overlapping candidates are resolved deterministically by retaining the longer valid span.

Operationally, the matcher first locates each match as a half-open interval of whitespace-delimited reference tokens and stores the corresponding character offsets. The formulation above uses the stored character interval because decoder subwords need not coincide with whitespace-token boundaries.

Let $[u_{i,t}, w_{i,t})$ denote the character interval associated with decoder token t after tokenization. The weak label for that position is

$$e_{i,t}^* = \begin{cases} c_{i,k}, & \text{if } \exists k \leq K_i : [u_{i,t}, w_{i,t}) \cap [a_{i,k}, b_{i,k}) \neq \emptyset, \\ O, & \text{otherwise.} \end{cases} \quad (4)$$

Special decoder tokens and padding are ignored. If a token intersects multiple retained spans, it receives the type with the greatest character overlap, with ties resolved in favor of the longer span. This yields token-aligned supervision while retaining the spoken-form transcript as the decoder target.

The matcher is applied only to the frozen train-side references used for the matched adaptations; development and test references never add patterns, phrases, or mappings. This scope prevents the evaluation references from becoming supervision. The labels are therefore weak supervision rather than manually verified named-entity annotations. Ambiguous matches are omitted instead of being assigned a guessed value. The frozen training labels retain all accepted matcher outputs, including 95 speed spans; development and test references are not used to add or remove entity types. Speed conversion is also supported by the restoration module.

5.3 Entity-Aware LoRA Adaptation

The weak labels are used to shape the recognizer rather than to introduce a second output format at inference. We use LoRA to

adapt a Whisper aviation recognizer [10, 23]. During teacher-forced training, for each decoder position t the model produces a hidden state $h_{i,t}$ and token distribution $p_{\theta}(z_{i,t} \mid x_i, z_{i,<t})$. Because $h_{i,t}$ combines the decoded history with cross-attended acoustic evidence, an auxiliary type objective can encourage the representation to retain entity-bearing information without changing the transcript vocabulary. A linear head predicts a class in C_0 :

$$p_{\psi}(e_{i,t} \mid h_{i,t}) = \text{softmax}(W_e h_{i,t} + b_e). \quad (5)$$

We denote the predicted type by $\hat{e}_{i,t} = \arg \max_{c \in C_0} p_{\psi}(c \mid h_{i,t})$. The transcription and entity losses are

$$\mathcal{L}_{\text{ASR}} = - \sum_{i,t} \log p_{\theta}(z_{i,t}^* \mid x_i, z_{i,<t}^*), \quad (6)$$

$$\mathcal{L}_{\text{ent}} = - \sum_{i,t} \omega(e_{i,t}^*) \log p_{\psi}(e_{i,t}^* \mid h_{i,t}), \quad (7)$$

where the sums run over valid decoder positions, special tokens and padding are ignored, $\omega(O) = 0.25$, and positive entity labels have unit weight. Downweighting the frequent non-entity class prevents it from dominating the auxiliary objective. The joint objective is

$$\mathcal{L} = \mathcal{L}_{\text{ASR}} + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}}, \quad \lambda_{\text{ent}} = 0.2. \quad (8)$$

The auxiliary classifier is used only to shape the shared decoder representation during adaptation. It is discarded at inference, so decoding follows the same autoregressive ASR path and output format as the matched ordinary LoRA control. Consequently, the comparison isolates the effect of structural supervision rather than introducing a larger inference model or a privileged entity list at decoding time.

5.4 Schema-Constrained Restoration Module

The restoration module $g_{\mathcal{K}}$ maps a spoken-form hypothesis to canonical written text in one externally visible module. Given $\hat{z}_i$, a detector produces a set $\mathcal{M}_{\mathcal{K}}(\hat{z}_i)$ of non-overlapping candidates. Each candidate $m = (\ell_m, r_m, c_m, \tilde{y}_m)$ contains a source span $[\ell_m, r_m)$, an entity type c_m , and a composed canonical form $\tilde{y}_m$. A deterministic gate $q_{\mathcal{K}}$ accepts a candidate only when its lexical context, digit length, and output format satisfy the corresponding structural contract. Thus,

$$\hat{y}_i = g_{\mathcal{K}}(\hat{z}_i) = \text{Apply}(\hat{z}_i, \{m \in \mathcal{M}_{\mathcal{K}}(\hat{z}_i) : q_{\mathcal{K}}(m, \hat{z}_i) = 1\}), \quad (9)$$

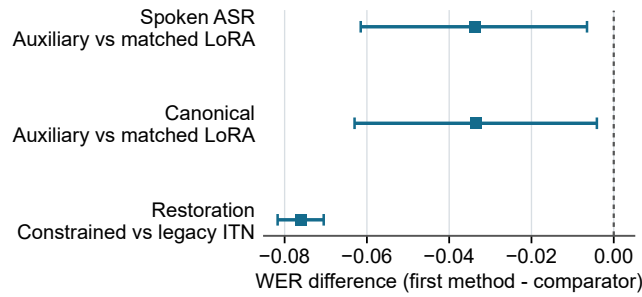
where Apply replaces accepted spans from left to right and copies every rejected or unmatched span unchanged. Supported operations include aviation digit and unit composition together with general spoken-to-written normalization. Examples include *zero two right*→02R, *flight level tree five zero*→FL350, and *one two one decimal seven*→121.7.

For spoken call signs, prefix-code associations are estimated from training-only paired text groups and retained only above fixed support and ambiguity thresholds. The mappings and all structural rules are frozen before test; exact thresholds are provided in the supplement. Local validation copies unsupported spans unchanged. It cannot recover evidence already lost by ASR, but prevents restoration from modifying content unsupported by the hypothesis.

Table 2: Results on 4,487 test recordings. Normalized sentence exact match and entity-value F1 are percentages; WER is corpus-level. Bold marks the best value among the displayed systems. The daggered row is exploratory.

Method	Spoken-form ASR			Legacy ITN		Schema-constrained restoration	
	Norm. exact (%) $\uparrow$	WER $\downarrow$	Entity F1 ‡ (%) $\uparrow$	Norm. exact (%) $\uparrow$	WER $\downarrow$	Norm. exact (%) $\uparrow$	WER $\downarrow$
XLS-R (ATCOSIM)	0.00	0.9862	5.64	0.00	1.0886	0.00	1.0879
XLS-R (UWB-ATCC+ATCOSIM)	0.25	0.8924	22.67	0.25	1.0224	0.25	1.0190
Whisper large-v3	22.64	0.6767	38.97	24.23	0.6529	25.21	0.6404
Whisper large-v3-turbo	21.53	0.6912	36.92	22.82	0.6810	23.78	0.6660
Whisper large-v3 ATC FT	14.93	0.6183	49.18	14.66	0.7269	15.47	0.6976
Whisper ATCO2 domain base	22.44	0.5679	52.43	22.69	0.6668	24.23	0.6328
+ matched LoRA	33.96	0.5593	67.11	29.66	0.6938	33.36	0.6192
+ entity-token weighting	34.28	0.5418	67.88	29.77	0.6755	33.63	0.5971
+ consolidated entity-auxiliary LoRA (ours)†	37.02	0.4980	65.29	32.23	0.6246	36.39	0.5568

‡ Computed on the 1,131 reference-entity-positive utterances. † Exploratory single-adaptor consolidation; protocol and adaptive-analysis disclosure are in the supplement.

**Figure 3: Prespecified corpus-WER differences on 4,487 test recordings with 95% paired-bootstrap intervals; negative favors the first method. Fusion is excluded from confirmatory inference.**

6 Experimental Setup

6.1 Controlled Comparison Setup

All controlled adaptations use the same 48,098 training recordings, domain base, decoder configuration, and fixed training-reference matcher. The matcher labels 17,076 spans without development or test input.

6.2 Models and Training

We initialize from jlvdoorn/whisper-large-v3-atco2-asr [12, 23, 35]. Two aviation XLS-R models [2] and Whisper large-v3 variants are off-the-shelf references; controlled comparisons use matched standard LoRA, entity-token weighting, and entity-auxiliary LoRA on the domain base.

Controlled runs use decoder-only rank-8 LoRA for 6,013 steps. Matched LoRA and Auxiliary use unit ASR token weights; weighting uses 1.5 on weak entity tokens, and Auxiliary uses $\lambda_{\text{ent}} = 0.2$ and $\omega(O) = 0.25$. Each variant is one fixed run; optimization and decoding details are in the supplement.

The exploratory consolidated variant is a single adapter with unchanged greedy decoding. Its exact construction, endpoints, inference cost, and adaptive-analysis disclosure are given in the supplement.

System	Output
Spoken reference	deviation from the route is limited within five kilometers on each side due to dangerous area
Whisper large-v3	divide from the road is limited within 5 kilometers on each side due to dangerous area
Whisper large-v3 ATC FT	deviation from zero is limited within one klm on each side due to dangerous area
Matched LoRA	deviation from the north is limited within five kilometers on each side due to dangerous area
Entity-auxiliary LoRA	deviation from the route is limited within five kilometers on each side due to dangerous area [exact]

Figure 4: Case study of different methods. Auxiliary exactly reproduces the spoken reference, whereas the displayed comparators substitute route-related words.

The deterministic aviation ITN baseline is fixed before evaluation and uses no evaluation-derived correction or sentence lookup. The constrained module adds the schema and train-derived associations in Section 5.

6.3 Evaluation Protocol

Hyperparameters and restoration rules for the confirmatory systems were selected on development data and fixed before test evaluation. The consolidated variant is reported as exploratory rather than confirmatory; the supplement documents its adaptive-analysis protocol.

Spoken-form metrics use z_i^* and canonical metrics use y_i^* . We report corpus WER, normalized sentence exact match, and schema-based entity-value F1 on 1,131 parsed-entity test utterances. Prompt-derived z_i^* measures expected-form agreement rather than verbatim accuracy; normalization and parser details are in the supplement.

We pre-specified three paired comparisons: Auxiliary versus matched LoRA, their outputs under the same restoration module, and constrained versus legacy restoration on the same recognizer. We use 20,000 paired bootstrap resamples, WER randomization, exact McNemar tests, and Holm correction within each metric family. The tests quantify fixed-checkpoint sample uncertainty, not training-run variation; consolidated-variant intervals are descriptive.

Table 3: Controlled adaptation comparison on 4,487 test recordings. Entity weight is the ASR-loss weight assigned to weakly labeled entity tokens; normalized exact match and entity F1 are percentages.

Variant	Entity weight	Aux.	Norm. exact $\uparrow$	WER $\downarrow$	Entity F1 $\uparrow$
Matched LoRA	1.0	no	33.96	0.5593	67.11
Token weighting	1.5	no	34.28	0.5418	67.88
Auxiliary	1.0	yes	33.56	0.5255	67.76

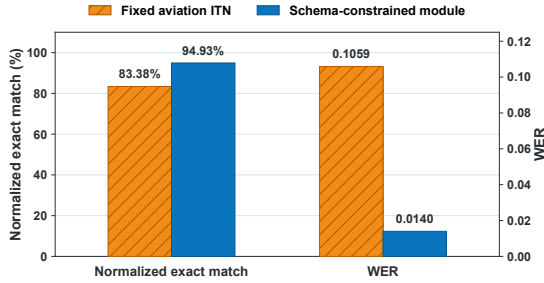


Figure 5: Restoration with prompt-reference input on all 5,922 development recordings. Identical prompt-derived spoken references exclude upstream ASR errors; normalized exact match and corpus WER use the left and right axes, respectively.

7 Results

7.1 Reference Systems and Controlled Adaptation

The off-the-shelf systems in Table 2 yield spoken-form WERs of 0.6183–0.9862 and are treated as descriptive references because they differ in architecture, training data, and output conventions. The ATCO2 domain base reduces WER to 0.5679 and obtains 52.43% entity F1. Applying matched LoRA increases normalized exact match from 22.44% to 33.96% and entity F1 from 52.43% to 67.11%, although WER decreases only to 0.5593. Domain adaptation therefore substantially improves sentence- and entity-level agreement, but its effect differs across metrics.

Figure 3 summarizes the prespecified comparisons. Relative to matched LoRA, Auxiliary reduces WER from 0.5593 to 0.5255, a 6.0% relative reduction (difference -0.0338 , 95% CI $[-0.0615, -0.0065]$; adjusted $p = 0.029$). Normalized exact match changes from 33.96% to 33.56% and is not significantly different ($p = 0.241$). Entity-token weighting instead obtains the highest exact match (34.28%) and entity F1 (67.88%), whereas Auxiliary obtains the lowest WER. This comparison indicates a metric-dependent trade-off between emphasizing weakly labeled tokens directly and using entity classification as an auxiliary representation objective.

Auxiliary achieves 89.86% precision, 54.39% recall, and 67.76% micro F1 on the 1,131 entity-positive utterances. Supplementary Table S9 shows that call signs remain the main recall bottleneck, with 48.13% recall and 61.60% F1, whereas distance and frequency reach 80.87% and 78.89% F1. These parser-based scores are structural diagnostics rather than independently annotated NER results.

7.2 Qualitative Case Analysis

Figure 4 compares recognition of a navigation-related utterance containing *five kilometers*. Generic Whisper, ATC-fine-tuned Whisper, and matched LoRA introduce different route- or distance-related substitutions, whereas Auxiliary alone reproduces the complete expected form. This is the sole Auxiliary-only exact case among the displayed systems in the frozen 50-recording development diagnostic, where Auxiliary obtains the lowest WER but entity-token weighting retains the highest entity F1 (Supplementary Table S8). The case therefore illustrates the observed metric trade-off rather than uniform superiority.

7.3 Exploratory Consolidated Variant

The consolidated entity-auxiliary variant represents two rank-8 updates as one rank-at-most-16 adapter with a single model forward pass. It achieves 37.02% normalized exact match and 0.4980 WER, improving these metrics by 3.45 points and 0.0275 relative to Auxiliary. However, entity F1 decreases from 67.76% to 65.29% because 48 call-sign true positives are lost. Consolidation therefore improves aggregate recognition while weakening call-sign recall. Because test behavior motivated this hypothesis, the result remains exploratory and is excluded from confirmatory inference.

7.4 Schema-Constrained Restoration

Schema-constrained restoration reduces canonical WER and increases normalized exact match across all systems in Table 2. On fixed Auxiliary hypotheses, it raises exact match from 28.91% to 32.85% and reduces WER from 0.6618 to 0.5857 (difference -0.0761 , 95% CI $[-0.0817, -0.0705]$; adjusted $p < 0.001$), producing 177 additional exact outputs with no fixed-ITN-only exact output. With the same module, Auxiliary also reduces canonical WER by 5.4% relative to matched LoRA (adjusted $p = 0.029$). Supplementary Table S6 shows that the largest gain occurs in multi-turn practice, where exact match increases from 1.00% to 19.33% and WER decreases from 0.8520 to 0.5960.

7.5 Restoration with Prompt-Reference Input

To isolate restoration from recognition errors, we replace ASR hypotheses with prompt-derived spoken-form references for all 5,922 development recordings. Figure 5 shows that the constrained module increases normalized exact match from 83.38% to 94.93% and reduces WER from 0.1059 to 0.0140, an 86.8% relative reduction. Supplementary Table S5 further reports 39 exact regressions for fixed aviation ITN and none for the constrained module. This diagnostic demonstrates broad canonicalization coverage when the required lexical evidence is available, but it is based on development references and does not replace the end-to-end test comparison.

8 Conclusion

We present a schema-guided framework combining weak-label entity-aware LoRA with conservative restoration. Auxiliary reduced prompt-reference WER, while restoration improved canonical WER over fixed aviation ITN. Together, the results indicate that a shared aviation schema can support both recognition adaptation and conservative canonicalization. The evidence remains limited to controlled course data and fixed checkpoints.

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